

**INSTITUTE OF ACTUARIES OF INDIA**

**EXAMINATIONS**

**21<sup>st</sup> November 2020**

**Subject CP3 – Communication Practice**

**Time allowed: 3 Hours 30 Minutes (09.30 – 13.00 Hours)**

**Total Marks: 100**

## Background information

You are an Actuary heading the Investment Department of a Life Insurance Company. Over the years your Company has been specialising in management of funds for employers who wish to have their fund based group Superannuation schemes administered externally. The contributions are determined by the Employer or the Scheme Trustees and could be regular or could be one-off contributions based on the financial position of the Employer. The employer can make claims to meet benefit payments as per the scheme rules.

The Company is in the process of designing a performance-linked bonus scheme for its fund managers. You are assisting with the drawing up of the framework. There have been some discussions on how the performance should be assessed.

Fund managers are allowed to take positions on individual stocks with an objective that they beat the benchmark index returns. It is possible that individual fund managers deviate from the benchmark as well as the weights of each stock in the benchmark. There is no other specification which is provided to the fund managers. The fund managers have no control over when the contributions are made into the fund or when the funds are withdrawn. However, Employers and Trustees do interact with the fund managers and inform them of any large contributions or withdrawals.

It is being debated whether performance should be assessed based on absolute returns or returns achieved on the funds in excess of benchmark returns and how should the performance be evaluated allowing for the risks the fund manager assumes. There have also been questions as to how the return itself should be measured. As the fund managers do not have control over the external cash flows you have proposed that the return be calculated using the Time Weighted Rate of Return. An alternative view is that fund managers can at least partially influence the timing of the cashflows and therefore performance should be measured calculating the Money Weighted Rate of Return (MWRR). The MWRR is also what the clients see and compare it with the earning rates that they would have assumed for calculating their pension liabilities.

The head of Human Resources who has also been part of these discussions has written to you as follows.

*“Dear Lata,*

*I agree that a fund manager’s performance-linked bonus should depend on the investment performance of the fund. But I can see that arriving at what is the return and whether that return is good enough is not quite straightforward. Could you please summarise for me with the help of some examples the different measures which are being talked about, why would they differ and how we would assess whether these returns are good enough so that we can draw up a framework which addresses all the parameters? Please also provide your recommendations.*

*Thanks  
Geetha”*

The returns on the different funds have been computed and a note on performance evaluation of different managers has been prepared.

## Note on measurement of performance of investment portfolios

### Measures of performance

Returns over the year of the funds would be calculated by using the following measures.

**Money- Weighted Rate of Return (MWRR)** or the Internal rate of return is the rate of return that would produce the total ending value of a portfolio given the amounts and timing of the cash flows. It gives greater weight to time periods where the portfolio has greater value than to periods where the portfolio has less value.

$$V_0 (1+i)^T + \sum_t C_t (1+i)^{(T-t)} = V_T$$

$V_0$  is the market value of the fund at the beginning of the period

$V_T$  is the market value of the fund at the end of the period

$C_t$  is the net cashflow into the fund (excluding investment proceeds) at time  $t$

$i$  is the money-weighted rate of return

**Time Weighted Rate of Return (TWRR)** measures the compound rate of growth of the money initially invested in the portfolio over a stated measurement period.

$$(1+i)^T = \frac{V_{t1}}{V_0} \times \frac{V_{t2}}{(V_{t1} + C_{t1})} \times \dots \times \frac{V_T}{(V_{tn} + C_{tn})}$$

$V_0$  is the market value of the fund at the beginning of the period

$V_T$  is the market value of the fund at the end of the period of  $T$

$C_{tr}$  is the net cash inflow, excluding investment proceeds, at time  $tr$

$V_{tr}$  is the market value of the fund at time  $tr$  just before the cashflow

The contributions into and withdrawals from the fund are not under the control of the investment manager. Those inflows and outflows may significantly influence the money- weighted rate of return. To isolate the effects of the fund manager's decisions, the measure of return should therefore not depend on the external cash flows. The TWRR is therefore a better measure of performance of the fund manager.

The fund managers though cannot control the inflows and outflows can be informed of significant contributions beforehand. Considering the fact that clients use the MWRR to plan for their liabilities the MWRR cannot be ignored completely. Therefore some weight can be assigned to the performance using this measure.

While the above measures are for assessing absolute returns it also needs to be seen how the fund managers perform considering the risk which has been taken on by the fund managers in order to maximize returns. The Employers and Scheme Trustees, all other things being equal, would prefer lower risk (less variability of returns) to higher risk (more variability of returns). They would be keen on maximising the return on their investments while simultaneously trying to minimise the risks. Therefore it is proposed to calculate a reward to risk ratio like the Sharpe Ratio which is the portfolio return divided by the portfolio standard deviation over the period. The higher the value of the reward-to-risk ratio, the better the risk-adjusted return—that is, the higher the return per unit of risk.

The Employers and Trustees would also like to compare the returns against the benchmark returns. Returns in excess of benchmark returns should be calculated and assigned a weight in the performance evaluation.

The calculations of the above metrics have been done for two portfolios for the previous year and a brief explanation of how the performance of the two fund managers stacks up against the metrics defined above has been provided in the following sections.

|           | Benchmark returns |                 |
|-----------|-------------------|-----------------|
|           | Nifty 100 Values  | Monthly returns |
| 1-Jan-19  | 10198             |                 |
| 31-Jan-19 | 10189             | -0.1%           |
| 28-Feb-19 | 10567             | 3.7%            |
| 31-Mar-19 | 10678             | 1.1%            |
| 30-Apr-19 | 10700             | 0.2%            |
| 31-May-19 | 10678             | -0.2%           |
| 30-Jun-19 | 10800             | 1.1%            |
| 31-Jul-19 | 10805             | 0.0%            |
| 31-Aug-19 | 10850             | 0.4%            |
| 30-Sep-19 | 10600             | -2.3%           |
| 31-Oct-19 | 10800             | 1.9%            |
| 30-Nov-19 | 10723             | -0.7%           |
| 31-Dec-19 | 10756             | 0.3%            |
|           | Annual return     | 5.5%            |

| Fund A                                                |                           | Amounts in Rs. Million    |                              |                             |                 |      |                                  |
|-------------------------------------------------------|---------------------------|---------------------------|------------------------------|-----------------------------|-----------------|------|----------------------------------|
| Date                                                  | Contributions to the fund | Withdrawals from the fund | Fund Value before cash flows | Fund Value after cash flows | Monthly returns |      | Fund Values for MWRR calculation |
| 1-Jan-19                                              |                           |                           | 15,988                       | 15,988                      |                 |      | (15,987)                         |
| 31-Jan-19                                             |                           |                           | 16,000                       | 16,000                      | 0.08%           |      |                                  |
| 28-Feb-19                                             |                           |                           | 16,593                       | 16,593                      | 3.71%           |      |                                  |
| 31-Mar-19                                             |                           |                           | 16,804                       | 16,804                      | 1.27%           |      |                                  |
| 30-Apr-19                                             |                           |                           | 16,857                       | 16,857                      | 0.32%           |      |                                  |
| 31-May-19                                             |                           |                           | 17,233                       | 17,233                      | 2.23%           |      |                                  |
| 30-Jun-19                                             |                           | 10                        | 17,238                       | 17,228                      | 0.03%           |      | 10                               |
| 31-Jul-19                                             |                           | 90                        | 17,403                       | 17,313                      | 1.02%           |      | 90                               |
| 31-Aug-19                                             |                           |                           | 17,503                       | 17,503                      | 1.10%           |      |                                  |
| 30-Sep-19                                             |                           |                           | 17,553                       | 17,553                      | 0.29%           |      |                                  |
| 31-Oct-19                                             |                           |                           | 18,593                       | 20,093                      | 5.92%           |      |                                  |
| 30-Nov-19                                             | 1,500                     |                           | 19,503                       | 19,503                      | -2.94%          |      | (1,500)                          |
| 31-Dec-19                                             |                           |                           | 18,770                       | 18,770                      | -3.76%          |      | 18,770                           |
|                                                       |                           |                           |                              | TWRR                        | 9.3%            | MWRR | 8.6%                             |
| Standard deviation of the portfolio (std.dev)         |                           |                           |                              |                             | 2.5%            |      |                                  |
| Risk free return (RFR)                                |                           |                           |                              |                             | 6%              |      |                                  |
| Sharpe ratio=(TWRR-RFR)/std.dev                       |                           |                           |                              |                             | 1.3             |      |                                  |
| Return in excess of benchmark (TWRR-Nifty 100 return) |                           |                           |                              |                             | 3.8%            |      |                                  |

| Fund B    |                           |                           |                              | Amounts in Rs. Million      |                 |      |                                  |
|-----------|---------------------------|---------------------------|------------------------------|-----------------------------|-----------------|------|----------------------------------|
|           | Contributions to the fund | Withdrawals from the fund | Fund Value before cash flows | Fund Value after cash flows | Monthly returns |      | Fund Values for MWRR calculation |
| 1-Jan-19  |                           |                           | 25,000                       | 25,000                      |                 |      | (25,000)                         |
| 31-Jan-19 |                           |                           | 25,890                       | 25,890                      | 3.56%           |      | 0                                |
| 28-Feb-19 |                           | 5,000                     | 27,000                       | 22,000                      | 4.29%           |      | 5,000                            |
| 31-Mar-19 |                           |                           | 21,736                       | 22,000                      | -1.20%          |      |                                  |
| 30-Apr-19 |                           |                           | 22,012                       | 22,012                      | 0.05%           |      |                                  |
| 31-May-19 |                           |                           | 22,134                       | 22,134                      | 0.55%           |      |                                  |
| 30-Jun-19 |                           | 50                        | 22,207                       | 22,157                      | 0.33%           |      | 50                               |
| 31-Jul-19 |                           |                           | 22,189                       | 22,189                      | 0.14%           |      | 0                                |
| 31-Aug-19 |                           |                           | 22,800                       | 22,800                      | 2.75%           |      |                                  |
| 30-Sep-19 |                           |                           | 22,845                       | 22,845                      | 0.20%           |      |                                  |
| 31-Oct-19 |                           |                           | 23,100                       | 23,100                      | 1.12%           |      | 0                                |
| 30-Nov-19 |                           |                           | 22,750                       | 22,750                      | -1.52%          |      |                                  |
| 31-Dec-19 |                           |                           | 22,356                       | 22,356                      | -1.73%          |      | 22,356                           |
|           |                           |                           |                              | TWRR                        | 8.7%            | MWRR | 11.6%                            |

|                                                       |      |
|-------------------------------------------------------|------|
| Standard deviation of the portfolio (std.dev)         | 1.8% |
| Risk free return (RFR)                                | 6%   |
| Sharpe ratio=(TWRR-RFR)/std.dev                       | 1.4  |
| Return in excess of benchmark (TWRR-Nifty 100 return) | 3.2% |

The risk free return is the yield Government bonds  
Then fund managers have to beat the Nifty 100 returns.

### Commentary

- Absolute performance of Fund A is higher than that of Fund B based on the TWRR.
- In Fund B there was a large withdrawal before returns fell in the February 2019. The returns were relatively higher in the earlier months and therefore the money weighted return is higher for Fund B as compared to Fund A.
- The Sharpe ratio which measures the excess of return over a risk free return per unit of risk is higher for Fund B as compared to Fund A, though not very significantly. It can be therefore said that the return which has been achieved by the Fund manager of Fund B is better as the returns are more stable.
- The excess of return over the benchmark is higher for fund A than for Fund B. Currently the benchmarks are the same.

### Recommendations

- Fund or portfolio returns should be calculated using the time-weighted rate of return method. Time-weighted rates of return are not distorted by cash flows, so they reflect the true performance of the fund or portfolio.
- The money- weighted return gives greater weight to time periods where the portfolio has greater value than to periods where the portfolio has less value. Considering that pension funds have an expectation of investment return which is higher than the ones which they have used for the valuation of their liabilities, and fund managers have some ability to anticipate external cashflows, this metric can also be added to the performance evaluation but with a lesser weight.

- Standard deviation is being used as the measure of investment return risk. The Sharpe ratio is an important reward-to-risk ratios that can be used to compare a portfolio's excess return with a measure of portfolio risk. A reward to risk metric is important in order to ensure that fund managers take reasonable risks and do not expose the funds to very risky investments. This is especially important as the fund managers have been given a broadly free mandate with no restrictions.
- Relative returns allow for the comparison of a fund's return with the return on the benchmark and is an important measure to understand relative performance.

Taking into these considerations it is recommended that performance relative to benchmark be assigned a weight of 50%, the reward to risk ratio a weight of 40% and the MWRR a weight of 10%.

### Instructions to candidates

You can base your answers on the performance evaluation note which has been prepared with the proposed metrics computed for two of the bigger funds and the recommendation in answering the questions. There is no need to perform any calculations. There is no word limit which has been prescribed.

### Questions

- Q. 1)** Draft an email to the head of HR answering her queries. Your email should give an overview of the two measures of returns, the need for the two measures, how outperformance is measured and how performance in evaluation allows for the risk taken and the recommendation. [90]
- Q. 2)** The email is to the head of HR. How did you ensure that the tone of the email is appropriate for a senior colleague? [2]
- Q. 3)** Explain how you summarized some of the numerical data to provide the required clarity without cluttering the email. [4]
- Q. 4)** Give two examples of jargon that you avoided or included but with appropriate explanations. [4]

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